

SAFETY DATA SHEET**1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

Product Name	Penguin Cement 203
Manufacturer Name	Sunstar Chemical (Thailand) Co.,Ltd.
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2. HAZARDS IDENTIFICATION**PHYSICAL HAZARDS**

Flammable liquids	Category 2
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HEALTH HAZARDS

Acute toxicity (Oral)	Category 5
Acute toxicity (Dermal)	Category 5
Acute toxicity (Inhalation: Gases)	Not applicable
Acute toxicity: (Inhalation: Vapours)	Category 5
Acute toxicity (Inhalation: Dusts and mists)	Category 5
Skin corrosion/irritation	Category 2
Serious eye damage/eye irritation	Category 2
Respiratory sensitization	Classification not possible
Skin sensitization	Category 1
Germ cell mutagenicity	Category 2
Carcinogenicity	Category 2
Reproductive toxicity	Category 2
Specific target organ toxicity - Single exposure	Category 1
Specific target organ toxicity - Repeated exposure	Category 1
Aspiration hazard	Classification not possible

ENVIRONMENTAL HAZARDS

Hazardous to the aquatic environment (Acute)	Not classified
Hazardous to the aquatic environment (Long-term)	Not classified
Hazardous to the ozone layer	Classification not possible

ALLOCATION OF LABEL ELEMENTS**Hazard symbol****Signal word**

Danger

Hazard statement

H225 Highly flammable liquid and vapour.
 H315 Causes skin irritation.
 H317 May cause an allergic skin reaction.
 H319 Causes serious eye irritation.
 H341 Suspected of causing genetic defects.
 H351 Suspected of causing cancer.
 H361 Suspected of damaging fertility or the unborn child.
 H370 Causes damage to organs.
 H372 Causes damage to organs through prolonged or repeated exposure.

Precautionary statements**Prevention**

- P201 Obtain special instructions before use.
- P202 Do not handle until all safety precautions have been read and understood.
- P210 Keep away from heat/sparks/open flames/hot surfaces. No smoking.
- P233 Keep container tightly closed.
- P240 Ground/bond container and receiving equipment.
- P241 Use explosion-proof electrical/ventilating/lighting/ equipment.
- P242 Use only non-sparking tools.
- P243 Take precautionary measures against static discharge.
- P260 Do not breathe dust/fume/gas/mist/vapours/spray.
- P261 Avoid breathing dust/fume/gas/mist/vapours/spray.
- P264 Wash thoroughly after handling.
- P270 Do not eat, drink or smoke when using this product.
- P272 Contaminated work clothing should not be allowed out of the workplace.
- P280 Wear protective gloves/protective clothing/eye protection/face protection.

Response

- P314 Get medical advice/attention if you feel unwell.
- P321 Specific treatment (see label).
- P302+P352 IF ON SKIN: Wash with plenty of soap and water.
- P308+P311 IF exposed or concerned: Call a POISON CENTER/doctor/...
- P308+P313 IF exposed or concerned: Get medical advice/attention.
- P332+P313 If skin irritation occurs: Get medical advice/attention.
- P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
- P337+P313 If eye irritation persists: Get medical advice/attention.
- P362+P364 Take off contaminated clothing and wash it before reuse.
- P370+P378 In case of fire: Use carbon dioxide to extinguish.
- P303+P361+P353 IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
- P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Storage

- P405 Store locked up.
- P403+P235 Store in a well-ventilated place. Keep cool.

Disposal

- P501 Dispose of contents/container to..... (in accordance with local/regional/national/international regulation).

3.COMPOSITION/INFORMATION ON INGREDIENTS

Substance/Preparation		Preparation
General Name		PVC type adhesive
Components Contributing to the Hazard		
Components	Cas No	%WT
Tetrahydrofuran	109-99-9	15-25
Acetone	67-64-1	50
Cyclohexanone	108-94-1	5-15

4.FIRST AND MEASURES**Inhalation**

Remove to fresh air and keep at rest in a position comfortable for breathing.

Skin Contact

Wash with plenty of soap and water.

Take off immediately all contaminated clothing. Rinse skin with water/shower.

If skin irritation or rash occurs: Get medical advice/attention.

Eye Contact

Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

Ingestion

Get medical attention immediately.

5.FIRE-FIGHTING

Extinguishing Media

Carbon dioxide, dry chemical, foam

Special Hazards

Toxic gases (chlorine compound) will form upon combustion.

Special Methods

Evacuate non essential personnel.

Move container from fire areas if it can be done without risk.

Keep personnel removed from and upwind of fire.

Protection of Firefighters

Wear self-contained breathing apparatus.

6.ACCIDENTAL RELEASE MEASURES

Personal Precautions

Wear protective equipment and avoid contact with skin.

Environmental Precautions

Do not allow to enter sewers/surface or ground water.

Methods for Cleaning up

Absorb with liquid-binding materials (sand, sawdust etc.).

Scoop material into containers for disposal.

7.HANDLING AND STORAGE

Handling

Technical Measures

Do not handle until all safety precautions have been read and understood.

Use only non-sparking tools.

Take precautionary measures against static discharge.

Wash thoroughly after handling.

Precautions

Do not breathe dust/fume/gas/mist/vapours/spray.

Do not eat, drink or smoke when using this product.

Use in the well - ventilated areas.

Storage

Suitable Storage Conditions

Keep away from heat/sparks/open flames/hot surfaces. No smoking.

Keep container tightly closed.

Store in a cool and well ventilated place.

Protect material from direct sunlight.

8.EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Components

Tetrahydrofuran

Acetone

Cyclohexanone

ACGIH TLV (2018)

50 ppm

250 ppm

20 ppm

Personal Protective Equipment

Hand Protection

Protective gloves

Eye Protection

Safety glasses

Skin and Body Protection

Avoid contact with skin.

Hygiene Measures

Wash hands before taking a break and at the end of work.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State

Form	Liquid
Color	Clear light yellow
Odor	Solvent odor
pH	—
Flashpoint	-11°C
Density	about 0.9
Solubility	Soluble in organic solvents
Other Data	—

10. STABILITY AND REACTIVITY

Stability

Stable under normal conditions.

Possible Hazardous Reactions

Reactive with water and amine compound.

Conditions to Avoid

Avoid water and amine compound.

Hazardous Decomposition Products

Chlorine compound

(in addition to CO , CO₂ , H₂O)

11. TOXICOLOGICAL INFORMATION

Acute toxicity

Tetrahydrofuran :	LD50 (Oral) :	Rat	1,650 mg/kg	(Ministry of the Environment Risk Assessment Volume 5: Preliminary Hazard Assessment Sheet (2006))
	LC50 (Inhalation: Vapours (4H)) :	Rat	18,187 ppm	(Ministry of the Environment Risk Assessment Volume 5: Preliminary Hazard Assessment Sheet (2006), ACGIH (7th, 2005), NTP TR475 (1998))
Acetone :	LD50 (Oral) :	Rat	5,800 mg/kg	(Ministry of the Environment Risk Assessment Volume 5: Preliminary Hazard Assessment Sheet (2008), SIDS (2002), ACGIH (7 th, 2001), EHC 207 (1998), ATSDR (1994))
	LD50 (Dermal) :	Rabbit	>7,400 mg/kg	(SIDS (2002))
Cyclohexanone :	LC50 (Inhalation: Vapours (4H)) :	Rat	32,000 ppm	(PATTY (6th, 2012), SIDS (2002), EHC 207 (1998), ATSDR (1994))
	LD50 (Oral) :	Rat	1,620 mg/kg	(SIDS (access on Apr, 2009), DFGOT vol. 10 (1998))
	LD50 (Dermal) :	Rabbit	947 mg/kg	(DFGOT (1998), PATTY (2001))
	LC50 (Inhalation: Vapours (4H)) :	Rat	2,450 ppm	(ACGIH (2003))
	LC50 (Inhalation: Dusts and mists (4H)) :	Rat	8,000 ppm	(ACGIH (2003))

Skin corrosion/irritation

Tetrahydrofuran :	Not classified
Acetone :	Not classified
Cyclohexanone :	Category 2

Serious eye damage/eye irritation

Tetrahydrofuran :	Category 2A
Acetone :	Category 2B
Cyclohexanone :	Category 2A

Skin sensitization

Acetone : Not classified
Cyclohexanone : Category 1

Germ cell mutagenicity

Cyclohexanone : Category 2

Carcinogenicity

Tetrahydrofuran : Category 2
Cyclohexanone : Not classified

Reproductive toxicity

Tetrahydrofuran : Category 2
Acetone : Category 2
Cyclohexanone : Category 2

Specific target organ toxicity - Single exposure

Tetrahydrofuran : Category 1 (Central nervous system),
Category 3 (Respiratory tract irritation, Narcotic effects)
Acetone : Category 3 (Respiratory tract irritation, Narcotic effects)
Cyclohexanone : Category 1 (Respiratory system)
Category 2 (Central nervous system)
Category 3 (Narcotic effects)

Specific target organ toxicity - Repeated exposure

Tetrahydrofuran : Category 1 (Central nervous system, Respiratory system, Liver)
Acetone : Category 1 (Central nervous system, Respiratory system, Gastrointestinal tract)
Cyclohexanone : Category 1 (Central nervous system, Bone)

12.ECOLOGICAL INFORMATION**Hazardous to the aquatic environment (Acute)**

Tetrahydrofuran :	LC50 (96H) :	Fish (Fathead minnow)	2,160 mg/L	(ECETOC TR 91, 2003)
Acetone :	LC50 (96H) :	Fish (Fathead minnow)	>100 mg/L	(EHC 207, 1998)
Cyclohexanone :	LC50 (24H) :	Crustacea (Daphnia magna)	800 mg/L	(SIDS, 1996)
	LC50 (96H) :	Fish (Fathead minnow)	527 mg/L	(SIDS, 1996, etc.)

Hazardous to the aquatic environment (Long-term)

Tetrahydrofuran :	NOEC (35-38days) :	Fish (Fathead minnow)	216 mg/L	(Ministry of the Environment Risk Assessment Vol.7, 2009)
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13.DISPORSAL CONSIDERATIONS**Waste from Residues**

Follow the all regulations in your country.

Contaminated Packaging

Follow the all regulations in your country.

14.TRANSPORT INFORMATION

UN Class 3
UN No 1133
Packing group II

Specific Precautionary Transport Measures and Conditions

Follow the all regulations in your country.

15.REGULATORY INFORMATION

Follow the all regulations in your country.

16.OTHER INFORMATION

The information herein is given in good faith, but no warranty, express or implied, is made.

The information presented herein is believed to be accurate but is not warranted. Recipients are advised to confirm in advance that the information is current, applicable and suitable to there circumstances.